

Creative Media Analyzer

Roadmap (v2.1)

Open-source repositioning: current state, what's shipped, what's blocked on what

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Deployment <https://media-analyzer-theta.vercel.app>

Repository github.com/theodoreheroldrajan-sketch/media-analyzer

1. Purpose of this document

This is the single forward-looking source of truth for the project. Every item the team has discussed but deferred lives here, organised by *what's blocking it from shipping now*. The structure exists to make the answer to "what should I work on next?" obvious.

Three companion documents remain canonical for their respective scopes:

- `methodology.pdf / ANALYSIS_METHODODOLOGY.md` - how the math works
- `PRD_v2.md` - what the product is and for whom
- `LIMITATIONS.md` - honest accounting of what the tool cannot do (design choices, methodological limits, stack constraints)

This roadmap does not duplicate them. It points to them where relevant.

2. Current state

Deployment. Production at `media-analyzer-theta.vercel.app`. Root URL redirects to `/demo`. Real-app pages reachable by direct URL for the operator. Portfolio embedding via `iframe` from `portfolio-website-zeta-one-76.vercel.app/analyser`.

Architecture.

- Next.js 16.2.6 (App Router, Turbopack) on Vercel Hobby
- TypeScript 5.9.3, React 19.2.6, Tailwind 4
- Supabase Postgres (eu-west-2) with 9 tables and a creatives storage bucket
- Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) via Anthropic SDK with forced `tool_use`
- Streaming NDJSON over Node.js runtime
- GitHub Actions CI gating every PR to main (tsc, eslint, next build)

Codebase health (as of this writing). TypeScript clean. ESLint clean (zero errors, zero warnings). All findings from feature audit and code hygiene audit either shipped or deliberately deferred with documentation. CI workflow gating against regressions.

3. What's now shipped (post-v1)

Items that landed since the roadmap v1 was written, in rough chronological order.

Review fixes (`REVIEW_FIXES.md`, six items).

- Fix 1 - Model stability indicator (green/yellow/red traffic light for N/predictor ratio in Pro mode)

- Fix 3 - Pre-registration of hypothesis variables (database-backed, real app + demo)
- Fix 4 - Gated trust score (floor sub-scores cannot be averaged into a misleadingly good composite); since reduced to 5 sub-scores after confidence column was dropped
- Fix 5 - "Patterns to investigate" framing replaces "Top findings"; noise-adjusted ranking on both insights panel and variable performance table
- Fix 6 - Bootstrap 95% confidence intervals on every Lite delta, displayed as error bars and numeric ranges
- Fix 2 - Deferred pending user test (see section 6.2)

Post-audit work.

- Snapshot data layer: dashboard filters by `is_latest = true`, upload route flips flags correctly, re-uploads handled without mixing data
- "Showing snapshot N of M" indicator on dashboard
- Trust score formula updated in methodology paper and `ANALYSIS_METHODOLOGY.md` to match shipped code
- `LIMITATIONS.md` expanded with audit-surfaced operational gaps
- Confidence column investigation documented in `POST_AUDIT_COMPLETION.md` appendix; column subsequently dropped in PR #16

Audit cycle 1 (feature/correctness). Two reviews - external (chat-based, documentation-only) and internal (Claude Code, code-grounded). Findings consolidated in `AUDIT_FINDINGS.md` and addressed via `POST_AUDIT_INSTRUCTIONS.md`.

Audit cycle 2 (code hygiene). First hygiene audit. Findings in `CODE_HYGIENE_FINDINGS_2026-05-15.md`. Cleanup PRs:

- PR #16: app-shell deletion, README model name, notes column drop, unused exports, magic-number extraction, env-var assertion fix, eslint config - 8 findings addressed
- PR #17: React 19 effect refactor (`useSyncExternalStore` for `localStorage` contexts), patch bumps, Prettier toolchain, editor config, GitHub Actions CI workflow - 5 findings addressed

Open-source repositioning (PR #20). `LICENSE` (MIT), `README`, `CONTRIBUTING`, `SECURITY`, `CODE_OF_CONDUCT` files. `.env.example`, `docs/self-hosting.md`, `schema export`. Deploy-to-Vercel button. `PRD`, `LIMITATIONS`, `Methodology`, and this roadmap reframed for open-source / public-iteration posture. Build journey audit files moved to `docs/build-journey/`.

Audit cycle 3 (post-repositioning self-audit, PRs #21-23). Comprehensive review against the open-source repo state; fixes shipped without waiting for real test data.

- PR #21: Anthropic model ID correction, Supabase storage dedup, auth warning fix, initial Vitest suite - 4 findings

- PR #22: withRetry (2 attempts, exponential backoff) and pMap concurrency pool (5 workers default, configurable via ANALYSIS_CONCURRENCY) on the Anthropic extraction path; streaming handler wrapped in try-catch-finally with a stream_error event; CSV row cap (MAX_ROWS = 10_000, returns 413); concurrency guard on /api/analysis POST (409 if an active run within the last 15 min exists); auto-cleanup of runs stuck in running for more than 15 min on dashboard load; dashboard metric query-param validation; stale-closure fix in analysis/page.tsx; npm test step added to CI workflow; module-scope pricing constants; .env.example extended with optional ANALYSIS_MODEL and ANALYSIS_CONCURRENCY - 9 fixes
- PR #23: xorshift32 PRNG extracted to shared src/lib/rng.ts and threaded as optional seed through bootstrap CIs (deterministic tests; production unchanged); 10 MB image upload cap enforced in src/lib/upload.ts; all-zero-metric CSV rows filtered server-side with skippedRows count in the response; LIMITATIONS.md updated to mark 4 more items as resolved - 4 fixes

4. Build journey (preserved from v1)

This section is historical and unchanged from the v1 roadmap. It documents how the project was built, not where it's going. The forward-looking content is in sections 5 onward.

- **Streamlit prototype (April 22-23, 2026).** Five commits. Single-page Streamlit app proving Claude vision with tool_use could replace manual creative tagging. Preserved in Legacy/ as reference.
- **Next.js shell and Supabase foundation (May 11).** Pivot to Next.js 16 App Router with sidebar-based nine-step stepper. Supabase project, nine-table schema with snapshot model on performance_rows.
- **Wizard steps 1-8 (May 11-12).** Eight PRs in roughly 24 hours, one per step.
- **Interactive demo (May 13).** /demo route with deterministically-generated fake creatives, componentized dashboard, Pro/Lite mode split, inline SVG chart types.
- **Access control lockdown (May 13).** Root redirects to demo; real-app pages reachable by direct URL only.
- **Portfolio documentation set (May 13).** PRD, methodology, roadmap PDFs written and embedded on portfolio site.
- **Review and audit cycles (May 14-16).** Two audits, six review fixes, two cleanup PRs. State documented in this roadmap.

5. What's next - overview

Pending work falls into four categories distinguished by *what's blocking it*. The categories matter because they imply different conditions for unblocking:

- **Engineering hygiene** - ship when convenient. No external dependency.
- **Waiting for the first real user** - gated on a real performance marketer using the tool and providing feedback.
- **Waiting for real data at scale** - gated on the first 100-plus creative real dataset arriving.
- **Multi-tenant fork territory** - only matters if someone takes the codebase beyond single-operator self-host into a multi-user product. Documented for completeness; none of these ship on this repo.

A documentation maintenance task is also pending and is described at the end.

6.1 Engineering hygiene (ship when convenient)

No external dependency. These are quality-of-life improvements the team can tackle whenever there's time.

Major package bumps. Three packages are multiple major versions behind:

- @types/node 20 - 25 (Node 22 LTS now widespread)
- eslint 9 - 10
- typescript 5 - 6

Each needs its own scoped PR with breaking-change review. Do not batch.

Mass Prettier reformat. Prettier toolchain is installed (PR #17) but the codebase hasn't been reformatted. Running `npm run format` will produce a large diff. Worth doing as a standalone PR with no other changes mixed in.

React 19 rule semantics follow-up. The `react-hooks/set-state-in-effect` rule is new in React 19. Several `eslint-disable-next-line` comments were added in PR #17 where the effect pattern was correct but the rule could not detect that. When the rule stabilizes, revisit those sites.

M3 empty catch blocks. Five `try { ... } catch (e) {}` blocks across `analysis/page.tsx`, `settings/page.tsx`, and `variables/page.tsx`. All have inline comments documenting the silent-fail pattern as intentional. Not a finding to fix; flagged here only so future audits know they've been considered.

6.2 Waiting for the first real user

These items are gated on a real operator using the tool with a real dataset at scale. Building them on assumptions before that conversation risks shipping the wrong design.

- **Full snapshot UI.** Data-layer shipped; UI deferred. When real, the user will have re-uploaded CSVs and will want to see history. The minimum extension is a "Recent uploads" list on the settings page. Bigger extensions - switching the current snapshot, comparing two snapshots, deleting old ones - depend on what the user actually asks for.

- **Pro UI translation OR persona description correction.** Persona in PRD says "comfortable in ad platforms but not Python or R." Pro insights panel references coefficients and p-values inline. The user-test conversation resolves this: if they read coefficients comfortably, the persona description is wrong (update PRD); if they don't, the Pro UI needs plain-English translation.
- **AI insights narration design.** Depends on the outcome above. If translation is the path, this is where the narration patterns get implemented.
- **Bootstrap CI performance benchmark.** Bootstrap is computationally trivial on Lite-scale datasets but has not been measured on a real 120-plus creative dataset on Vercel Hobby. Benchmark with real data when available.

6.3 Waiting for real data at scale

These items are gated on the first 100-plus creative real dataset arriving. Methodological choices depend on the empirical structure of real ad-performance data, not synthesized data.

- **Production OLS backend.** Pro coefficient table remains mocked. Real backend builds against the first real dataset, with empirical decisions on regularization, impression weighting (count vs sqrt), and interaction selection threshold.
- **Personal knowledge-base layer.** Markdown notes (Obsidian-compatible) the operator maintains per brand. Notes get injected into extraction prompts so repeated campaigns inherit context. Designing this well needs to see what real campaign continuity looks like.
- **Longitudinal / temporal analysis (v3 direction).** Cross-snapshot analysis - creative fatigue, seasonality, pre/post brand campaigns, platform algorithm shifts. Gated on the first real user accumulating 3-plus snapshots over a 60-day window.

6.4 Multi-tenant fork territory

These only matter if someone wants to take the codebase beyond single-operator self-host into a multi-user product. Documented for completeness; none of these ship on this repo. If you fork to build a multi-tenant version, this is what you'd need to add.

- **Authentication and authorisation.** Currently UUID-based access. Anyone with a project URL has full destructive permissions. A multi-tenant version needs proper auth (Supabase Auth, RLS policies, role-based permissions).
- **Multi-user collaboration.** Shared workspaces, comments, audit trails, role-based access.
- **Video creatives.** Frame extraction, motion variables, audio extraction. Hybrid pathway with marketer-input variables for things AI cannot see (talent identity, music choice, production type). Probably v3 or later if anyone tackles it.
- **Real-time platform API integration.** Replaces CSV uploads with direct Meta/Google API integration. Requires credential management beyond current scope.

- **Advanced modelling.** Mixed-effects models for campaign-nested data. Bayesian regression for small-sample uncertainty. Tree-based methods (random forests, gradient boosting) with SHAP for interpretability. Only worth building if real data justifies the complexity.
- **Operational hardening.** The single-operator baseline (retry on transient Anthropic failures, concurrency guard, stuck-run cleanup, 10 MB image cap, CSV all-zero-row rejection) shipped in PRs #21-23 - see section 3. What remains here is the multi-tenant baseline: per-tenant rate limits, request quotas, audit logging of extraction runs, billing-grade cost telemetry per project, and tenant-scoped concurrency pools rather than the current process-global pool.
- **Resumable extraction.** Current practical batch ceiling is ~150-200 creatives per run on Vercel Hobby. Batching and resumable runs become necessary above that.
- **Accessibility audit.** Current dashboard relies on colour heavily (traffic-light indicators, sub-score bars). A multi-tenant product for diverse audiences needs a proper accessibility pass. Worth doing on this repo too if anyone wants to contribute it; the colour-only cues are a real issue regardless of tenancy.

7. Documentation maintenance

Regenerate the PDFs. The methodology, PRD, and roadmap markdown sources have been updated multiple times across the recent work cycles. The script `scripts/build_portfolio_docs.py` regenerates them. Run after any source markdown change that affects published content. This document (roadmap v2.1) is the latest run.

8. Pointers to canonical references

For specific topics, the source-of-truth document is:

- **What the tool does** - `PRD_v2.md`
- **How the math works** - `methodology.pdf` / `ANALYSIS_METHODOLOGY.md`
- **What the tool can't do** - `LIMITATIONS.md`
- **How to self-host** - `docs/self-hosting.md`
- **How to contribute** - `CONTRIBUTING.md`
- **How fixes were specified** - `docs/build-journey/REVIEW_FIXES.md`
- **Findings from feature audit** - `docs/build-journey/AUDIT_FINDINGS.md`
- **Post-audit work performed** - `docs/build-journey/POST_AUDIT_COMPLETION.md`
- **Confidence column investigation** - `docs/build-journey/POST_AUDIT_COMPLETION.md` (appendix)

- **Findings from code hygiene audit** - docs/build-journey/CODE_HYGIENE_FINDINGS_YYYY-MM-DD.md (latest dated file)
- **How to run audits** - docs/build-journey/AUDIT_INSTRUCTIONS.md, docs/build-journey/CODE_HYGIENE_AUDIT.md
- **Original build journey** - section 4 above

9. Open strategic questions

Decisions deferred until there is evidence to resolve them, not until "later." Listed so future-Theo (and any contributor reading) does not forget they are open.

- **First real-data engagement.** How to source the first real 100-plus creative dataset for production OLS validation. Options: wait for someone to clone the repo and offer it, actively recruit performance marketers willing to share anonymised data, or simulate from public benchmarks (probably worse than waiting). Open to suggestions.
- **Methodology review channel.** Whether to publish a short call-for-review post targeting people with applied-stats backgrounds, or rely on the README + GitHub Discussions to surface reviewers organically.
- **Paid media vs organic social.** Current functionality is paid-ads-shaped. Organic social would need a different metric set (saves, shares, comment sentiment) and probably a different schema. Both remain open.
- **Hybrid manual-input variables for video and image.** Custom variables exist but route through Claude vision. A future architecture extension would let users mark a variable as human-tagged (filled in by the marketer rather than extracted by AI). Useful for video where vision extraction is unreliable, and for image variables AI cannot see (talent identity, designer, internal A/B labels). Not committed.

10. Questions seeking input

Items where feedback from anyone reading the doc would help.

- **Personal knowledge-base layer.** Preferred scope and timing relative to production regression backend.
- **Pro vs Lite naming.** Whether the split should keep this naming if someone forks for a commercial product, or rename to something less paid-SaaS-coded.
- **Legacy folder.** Whether the legacy/ folder (Streamlit prototype) should be removed before the next public-facing iteration or retained as build-journey context. Currently retained.

- **Roadmap cadence.** Whether to publish dated milestone posts on the portfolio site or in GitHub Discussions tracking what shipped each week. Useful for build-in-public visibility, possible time sink. Open to opinions.